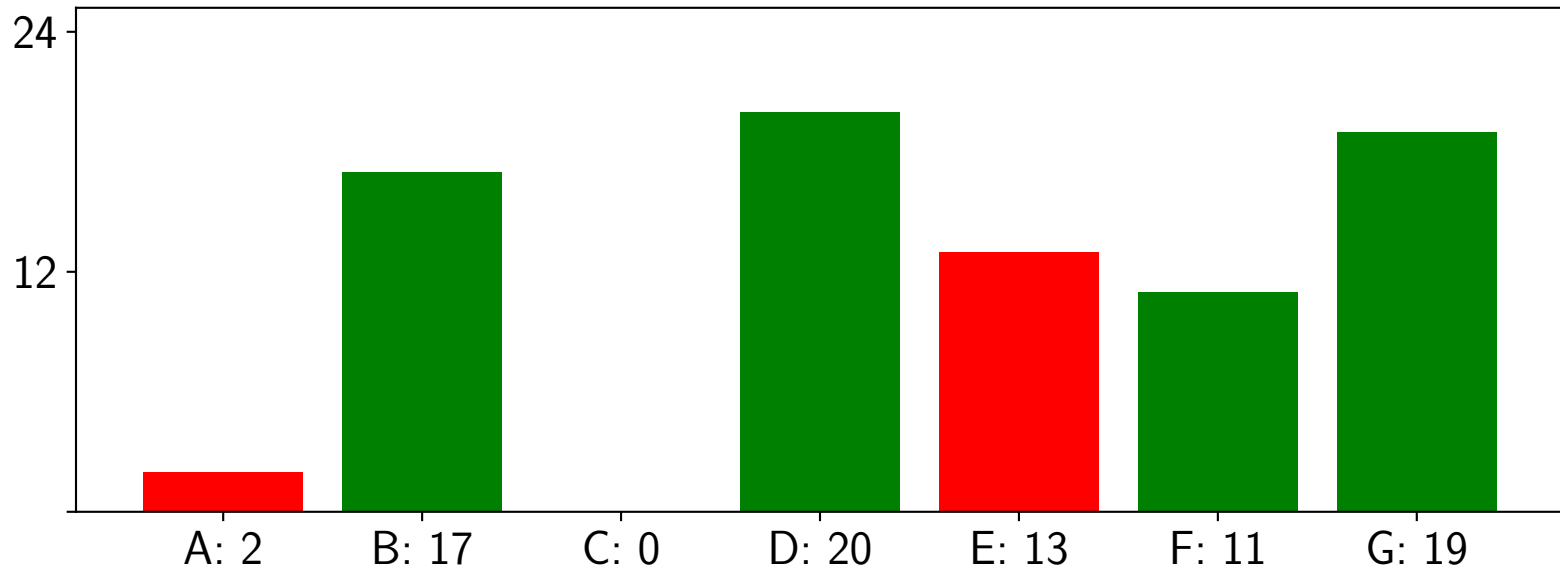


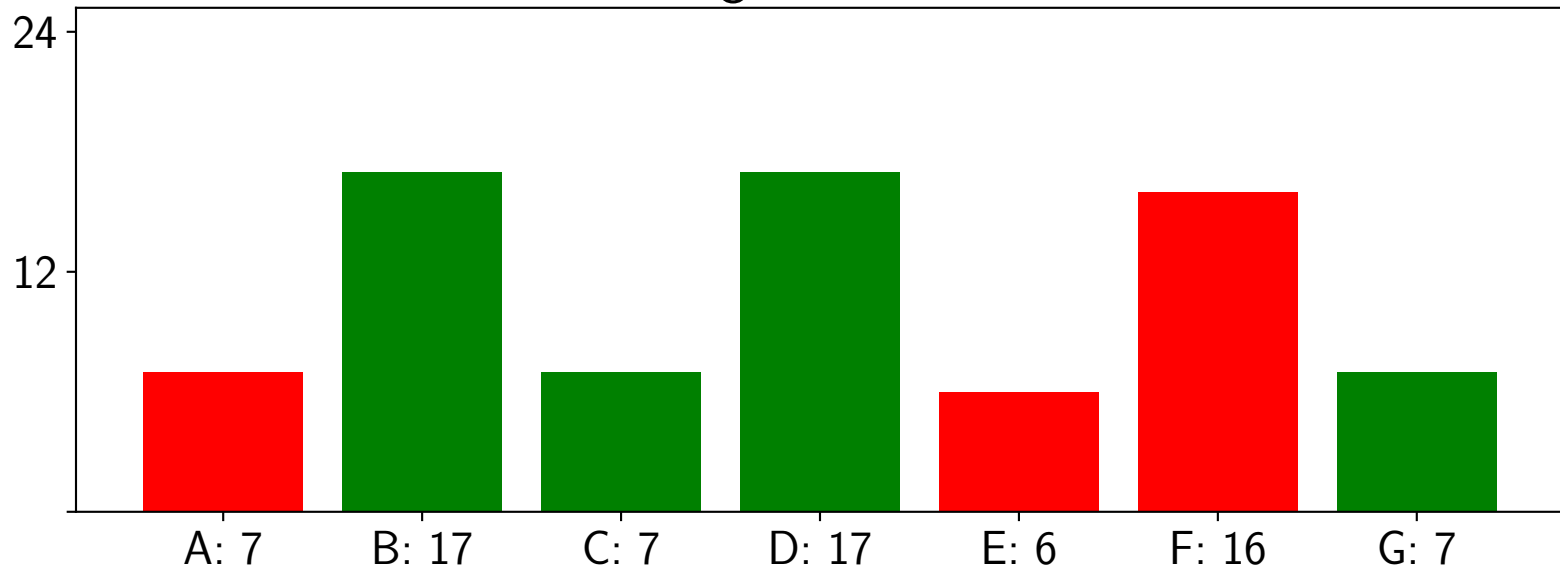
Convolutional Networks



Which statements about convolutional networks are correct?

- A Convolutional networks only contain operations related to convolution.
- B Stacked convolutional layers can increase the size of the receptive field.
- C Convolutional networks can only be used to process images.
- D More layers with smaller kernels are usually preferred over fewer layers with larger kernels.
- E Stacking many convolutional layers typically increases the performance of the network.
- F Rectified Linear Units (ReLU) and their variants help to speed up the training process.
- G Residual connections allow to train deeper networks.

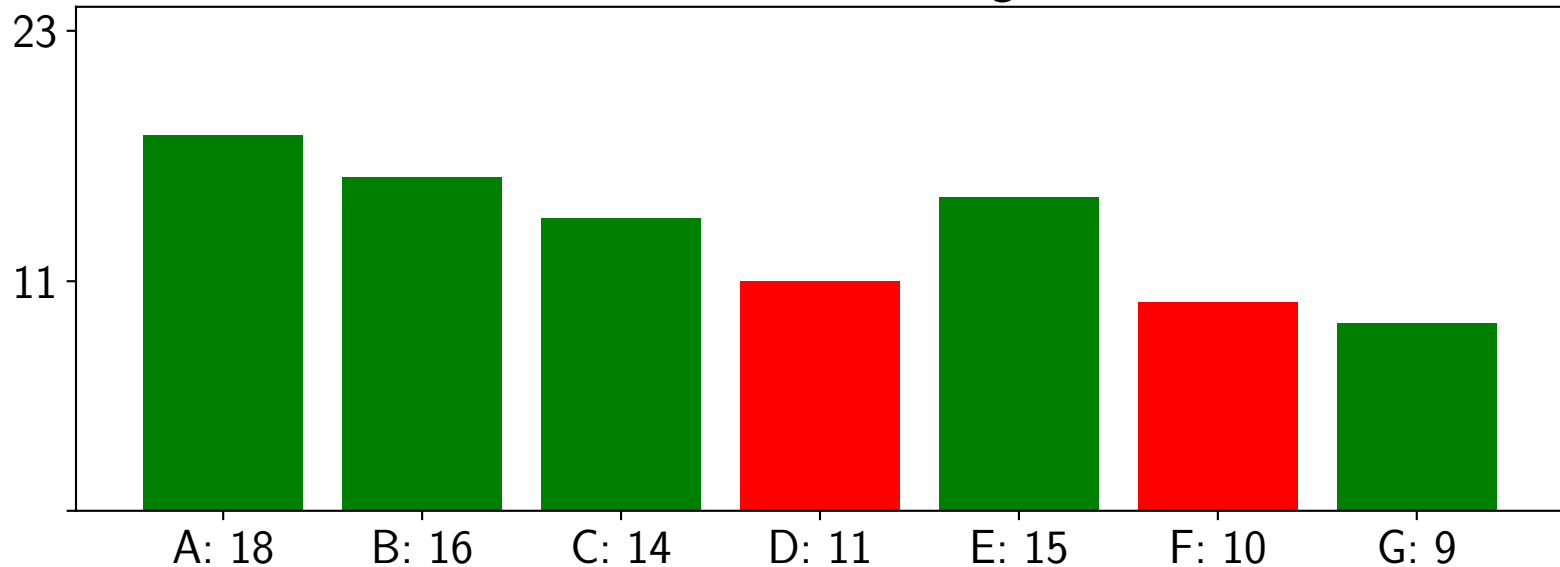
Regularization



Which of these techniques can be used to fight against the vanishing gradient problem?

- A Weight decay
- B Rectified Linear Units (ReLU)
- C Shortcut connections
- D Residual connections
- E DropOut
- F Batch normalization
- G Adam optimizer

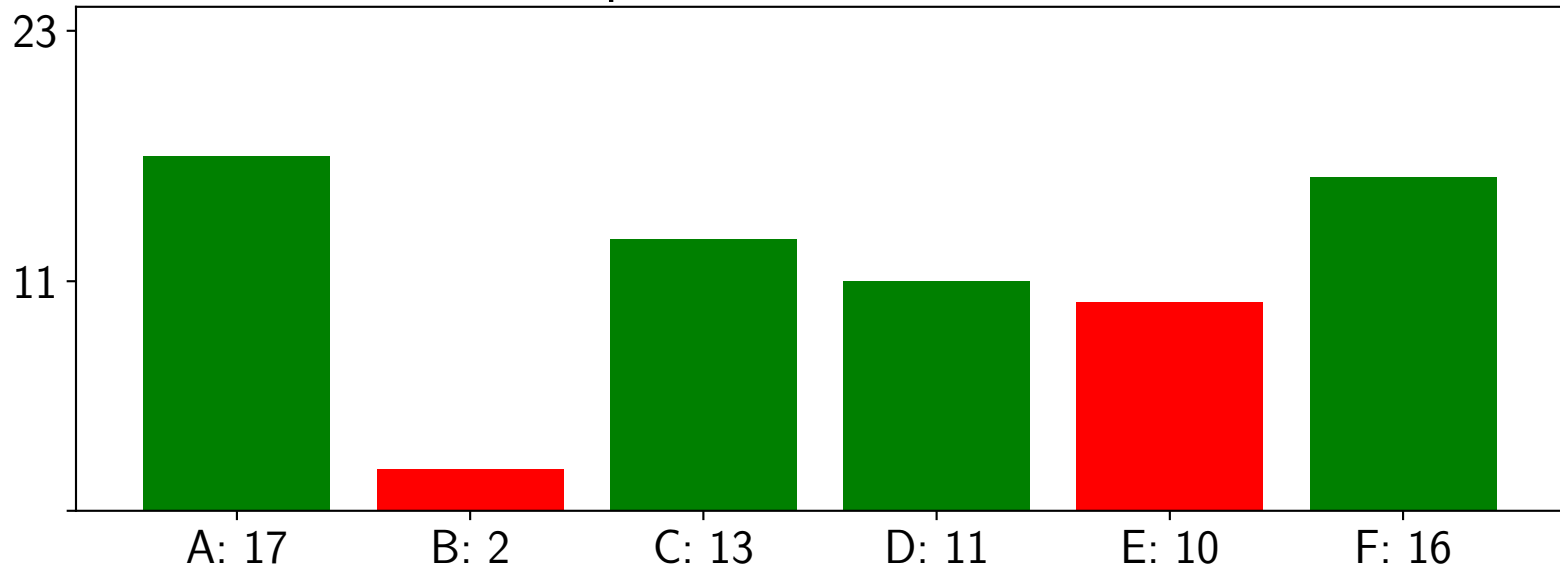
Transfer Learning



Which statements of transfer learning do you think are correct?

- A Transfer learning makes use of large training datasets to pre-train a deep network.
- B Domain adaptation fine-tunes a pre-trained network on a smaller target dataset.
- C Representation learning extracts deep features by removing the last fully-connected layer from the network.
- D Deep features are generally better than hand-tuned features.
- E Face recognition is a typical example for a representation learning task.
- F Pre-training a network with SoftMax loss provides well-discriminative features.
- G Deep features extracted from a deep network allow to distinguish known from unknown classes.

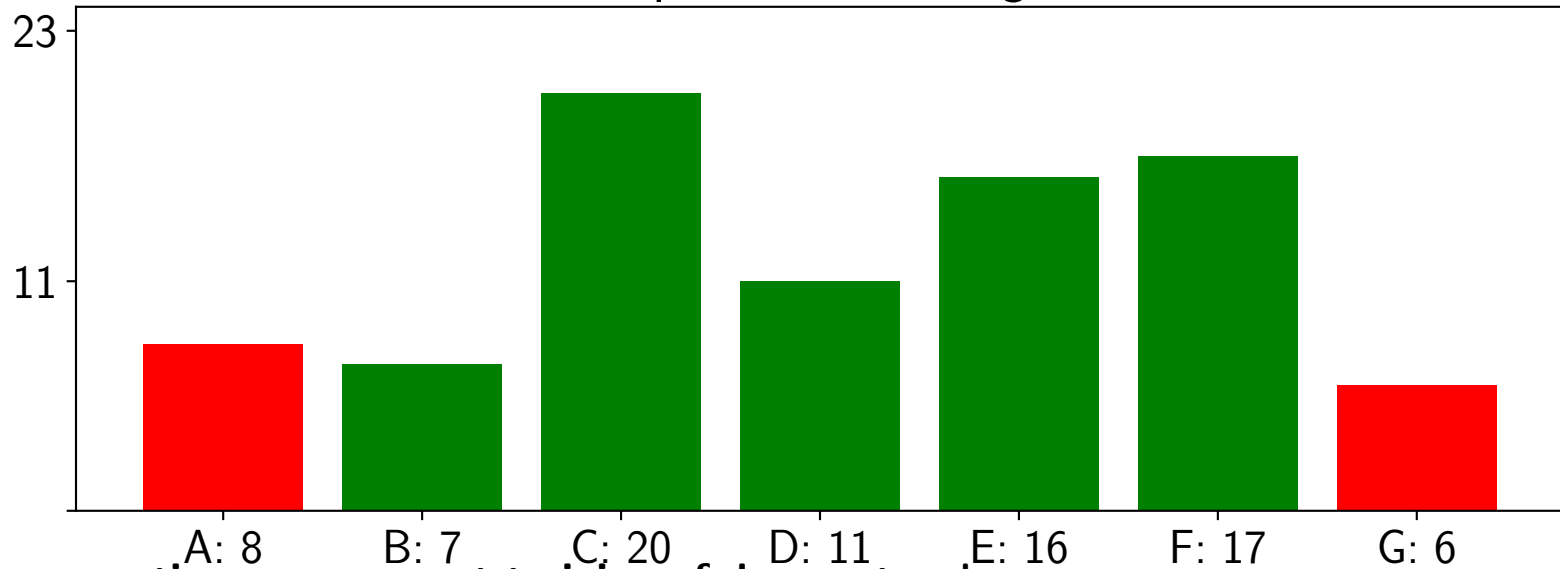
Deep Feature Visualization



Which statements about deep feature visualization are correct?

- A A bottle-neck network with two deep features can be used for visualization.
- B The bottleneck needs to be in the last layer of the network.
- C A fully-connected layer without bias neuron is applied on top of the deep features.
- D The last fully-connected layer can be interpreted geometrically.
- E An arbitrary number of classes can be visualized with the bottleneck network.
- F Each data sample builds one single point in the 2D flower plot.

Open-Set Training



Some questions on open-set training of deep networks.

- A All open-set classification techniques require negative samples during training.
- B When training only on the known classes, the softmax probabilities can be used to reject unknown samples.
- C Open-set networks can be trained by putting all negative samples into the same *Garbage* class.
- D The garbage class approach is used in most applications that need to handle unknown inputs.
- E Training with equal target values for negative samples pushes them to the center of the deep feature space.
- F Unknown samples typically end up in the same part of the feature space as the negative samples used during training.
- G Precision and Recall should be used to evaluate open-set classification tasks.